IBM Applied Data Science Capstone

Winning Space Race

with Data Science

Falcon 9 First-Stage Landing Prediction

Data collection • EDA • SQL • Folium • Plotly Dash • Classification

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Final Project Report

Executive Summary

Launch records	Landing success	Top launch site	Best classifier
90	66.7%	KSC LC 39A	Decision Tree
Falcon 9 missions	60 successful / 90	77.3% success rate	87.5% CV / 83.3% test

- Collected launch data through the SpaceX REST API and web-scraped launch records.
- Cleaned the data, handled missing values, and engineered a binary landing-success Class label.
- Used Python visualization and SQL to identify relationships among flight number, payload, launch site and orbit.
- Added interactive analytics with Folium and Plotly Dash, then compared four tuned classification models.
- Landing success increased strongly with flight experience; the Decision Tree led cross-validation (87.50%) with all four models achieving 83.33% test accuracy.

Introduction & Business Problem

Why landing prediction matters

- Falcon 9 reusability is a major contributor to SpaceX's lower launch economics.
- A competing launch provider needs to estimate recovery probability before bidding on a mission.
- The analytical question: given launch characteristics, can we predict whether the first stage will land successfully?

Questions addressed

- How does landing success change with flight number?
- Which launch sites and orbit types have higher success rates?
- Does payload mass show a meaningful relationship with recovery?
- Which classification model performs best?

End-to-End Methodology

1

Collect

SpaceX REST API
+ web scraping

2

Wrangle

Clean, filter,
engineer Class

3

Explore

Pandas/Matplotlib
+ SQL

4

Visualize

Folium
+ Plotly Dash

5

Predict

LR • SVM
• Tree • KNN

6

Decide

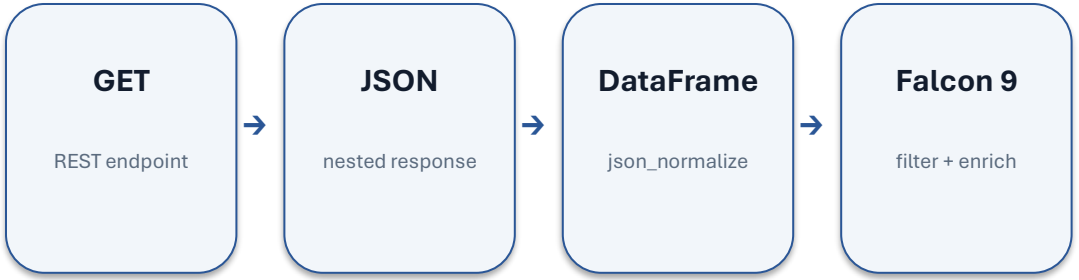
Compare metrics
& select model

Output: a stakeholder-oriented prediction of Falcon 9 first-stage landing success.

Data Collection — SpaceX REST API

API workflow

- GET historical launch records from the SpaceX REST API.
- Normalize nested JSON into a pandas DataFrame.
- Filter to Falcon 9 launches and enrich launch-site, payload and booster fields.
- Standardize missing payload values for downstream analysis.

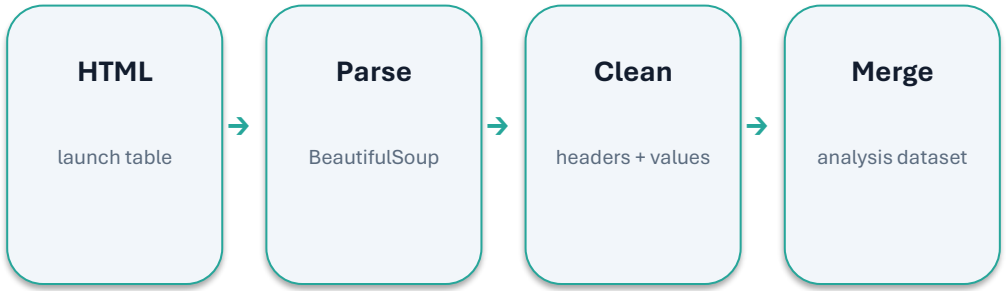


Resulting analysis table: 90 Falcon 9 launch records × 17 fields.

Data Collection — Web Scraping

Wikipedia launch-record extraction

- Retrieve the Falcon 9 launch-history table from Wikipedia.
- Parse HTML tables with Requests and BeautifulSoup.
- Normalize column headers, dates, payload units and landing-outcome labels.
- Use the scraped records to complement and validate API-derived fields.



Complementary sources reduce gaps and provide a richer launch-history view.

Data Wrangling

- Filtered the launch history to Falcon 9 missions.
- Reviewed missing values; LandingPad contains the principal missingness in the course dataset.
- Created the binary Class target: 1 = successful first-stage landing, 0 = unsuccessful / no recovery.
- Prepared numerical and categorical predictors for machine-learning models using feature selection and encoding.
- Final modeling dataset retains 90 observations with a 66.67% positive landing rate.

Records

90

Falcon 9

Success

60

Class = 1

Failure

30

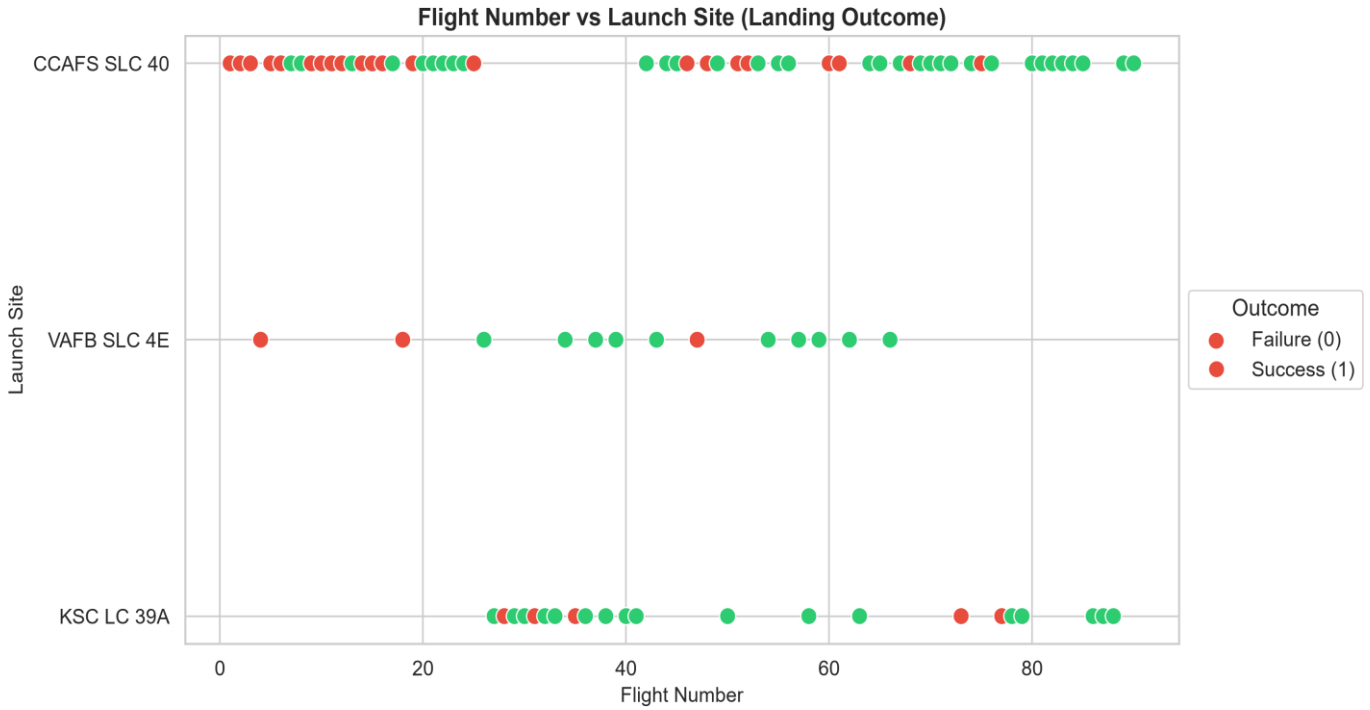
Class = 0

Baseline

66.7%

landing success

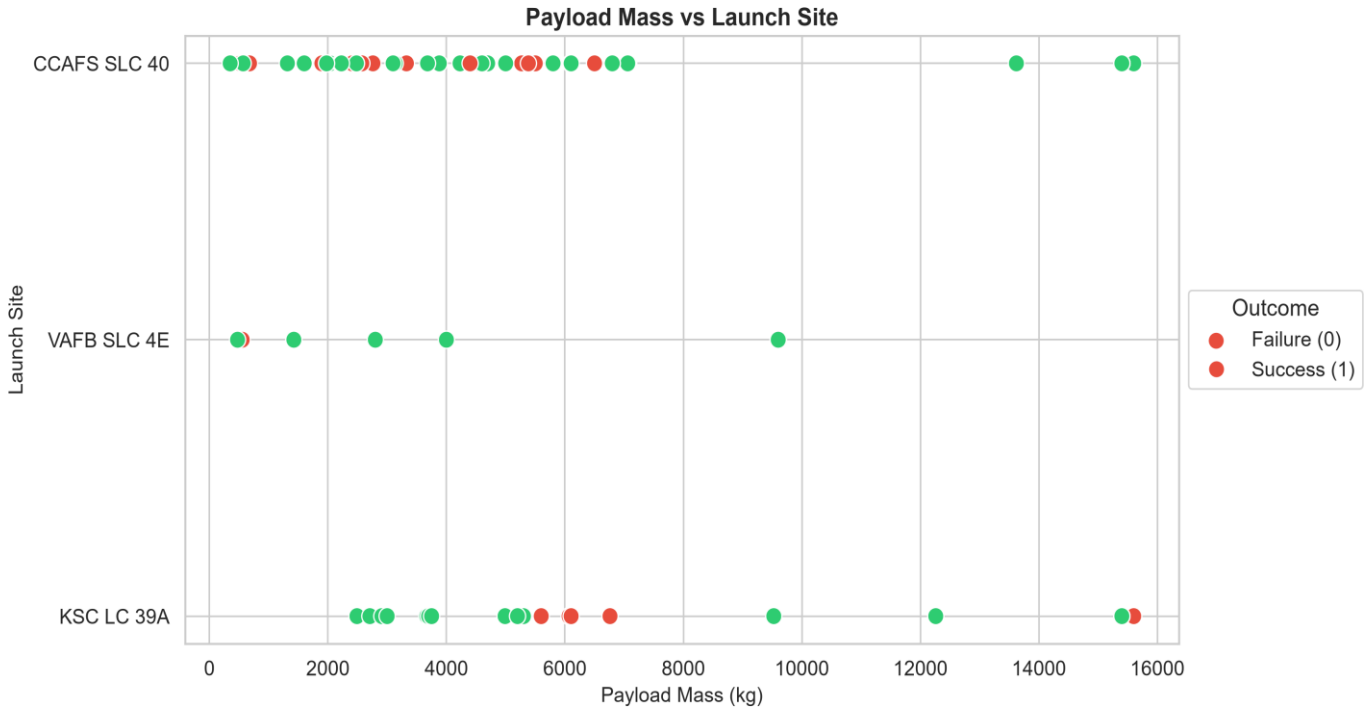
Flight Number vs. Launch Site



- CCAFS SLC 40 dominates the launch history across the full timeline.
- KSC LC 39A enters later and becomes a major site.
- Early flight numbers contain more failures; later flights cluster strongly toward successful landings.

EDA objective: identify variables and patterns that are useful for prediction and business interpretation.

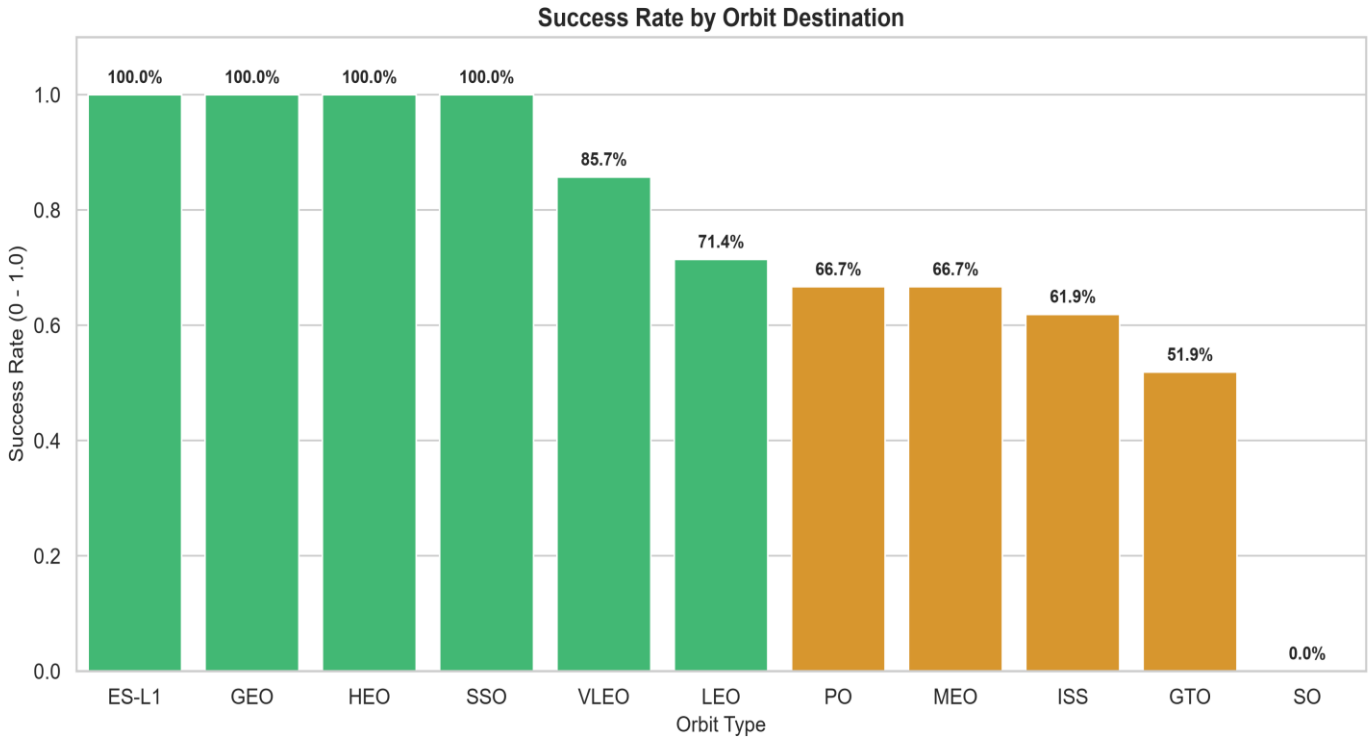
Payload Mass vs. Launch Site



- CCAFS SLC 40 and KSC LC 39A support a broad payload range.
- VAFB SLC 4E appears less frequently and is concentrated in lighter/specialized missions.
- High payload mass alone does not prevent successful recovery in later missions.

EDA objective: identify variables and patterns that are useful for prediction and business interpretation.

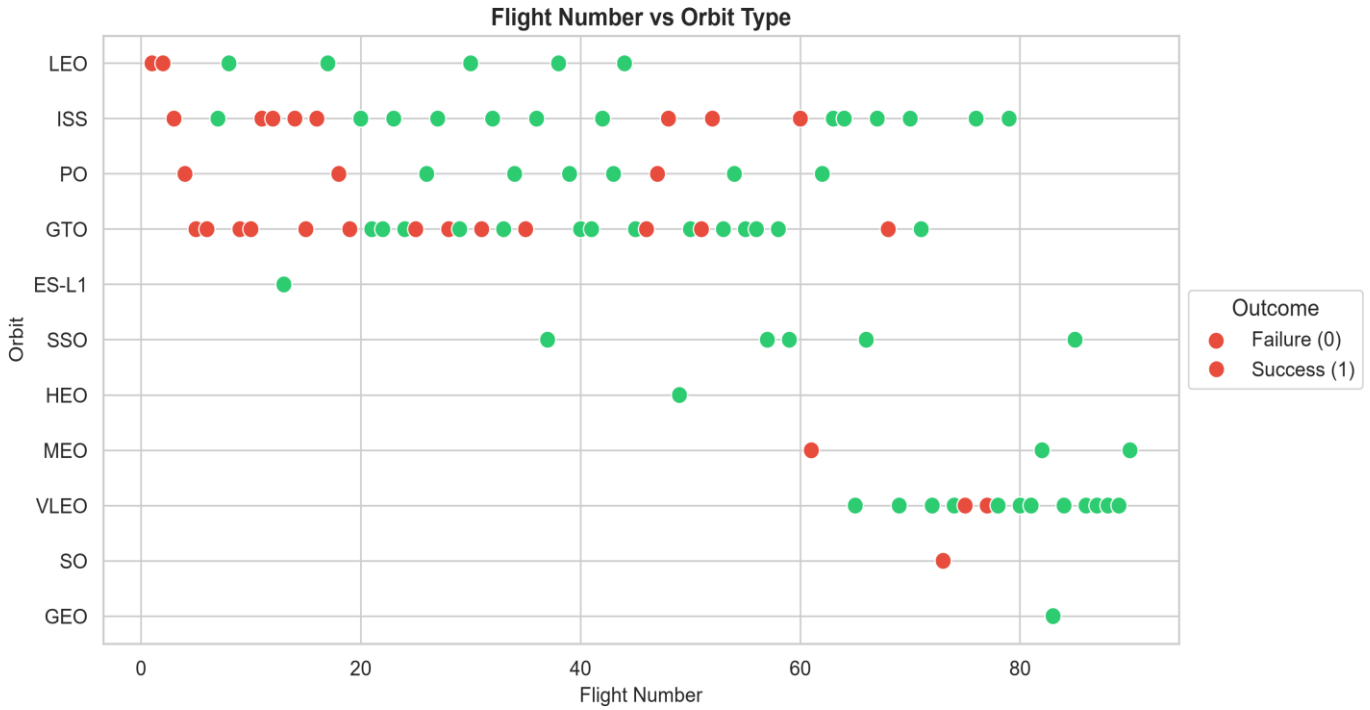
Success Rate vs. Orbit Type



- VLEO reaches 85.7% across 14 missions.
- SSO, GEO, ES-L1 and HEO show 100% in this dataset, but each has very few observations.
- GTO is the most common orbit and has the lowest success rate among the major mission groups at 51.9%.

EDA objective: identify variables and patterns that are useful for prediction and business interpretation.

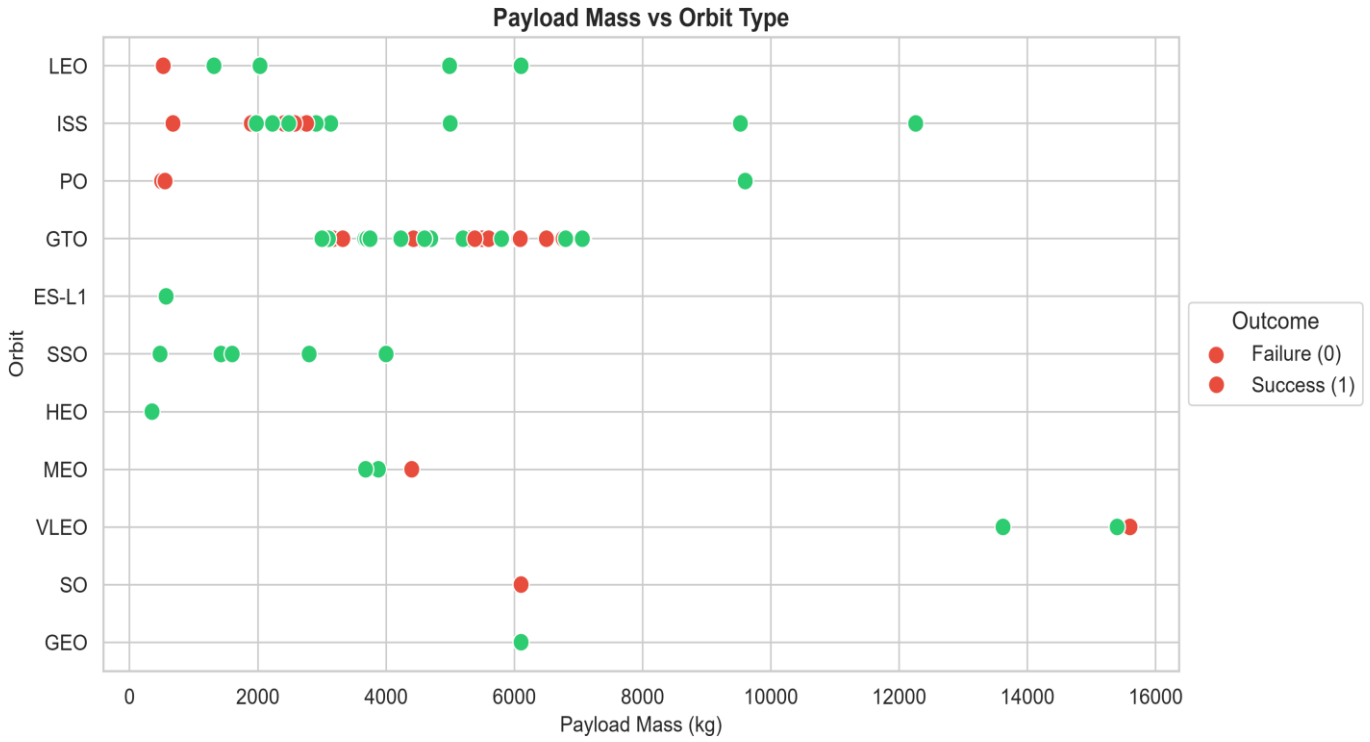
Flight Number vs. Orbit Type



- GTO and ISS are concentrated throughout the middle of the program.
- VLEO appears mainly in later flight numbers and is associated with high recovery success.
- Specialized orbits such as GEO, HEO and SO occur only rarely.

EDA objective: identify variables and patterns that are useful for prediction and business interpretation.

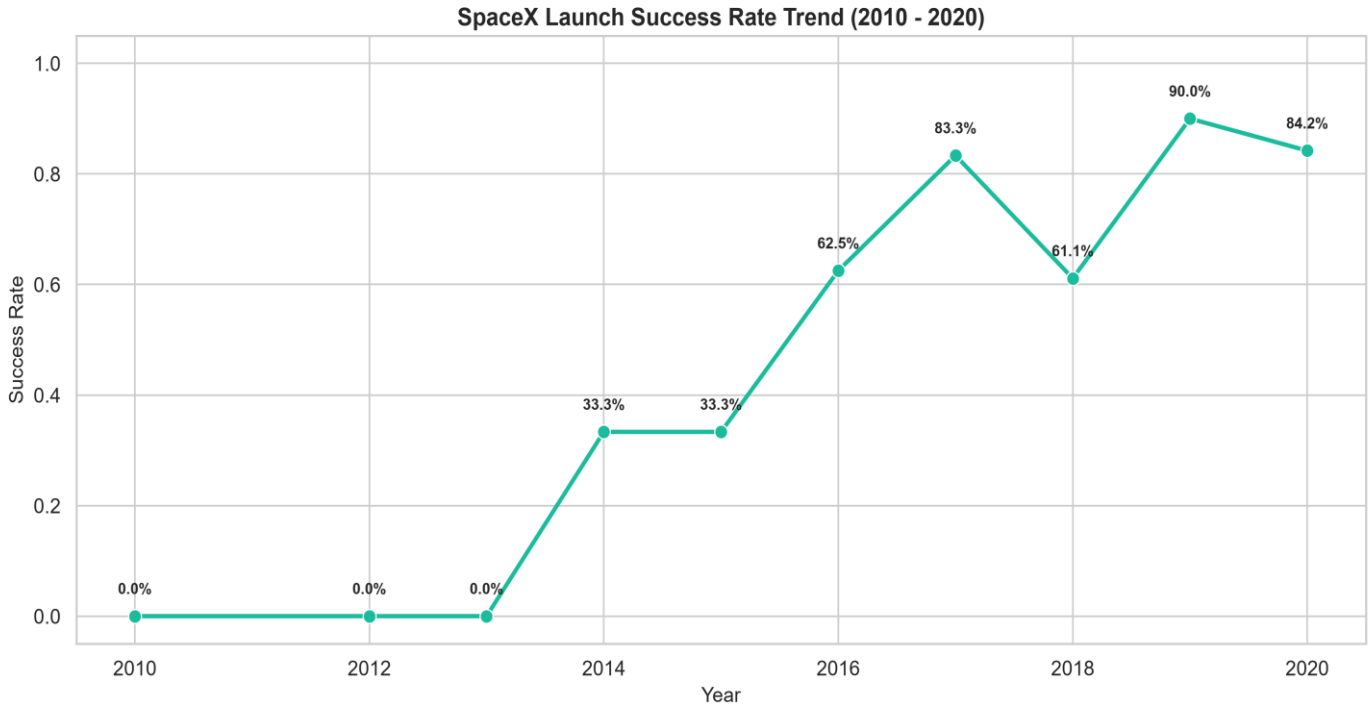
Payload Mass vs. Orbit Type



- Orbit type strongly constrains payload ranges.
- VLEO missions carry some of the heaviest payloads in the dataset while maintaining high recovery rates.
- Outcome patterns become more favorable as flight experience increases.

EDA objective: identify variables and patterns that are useful for prediction and business interpretation.

Launch Success Yearly Trend



- Success rises from 0% in the earliest years to more than 80% in 2017 and later.
- The 2019–2020 period remains above 80% in this dataset.
- The trend is consistent with improving booster-recovery maturity over time.

EDA objective: identify variables and patterns that are useful for prediction and business interpretation.

EDA with SQL — Key Query Results

Unique launch-site labels

4

CCA... LC-40 / SLC-40, KSC LC-39A, VAFB SLC-4E

NASA (CRS) payload

45,596 kg

SUM(payload_mass__kg_)

F9 v1.1 average payload

2,928.4 kg

Exact F9 v1.1 (2,534.7 kg with sub-variants)

First ground-pad success

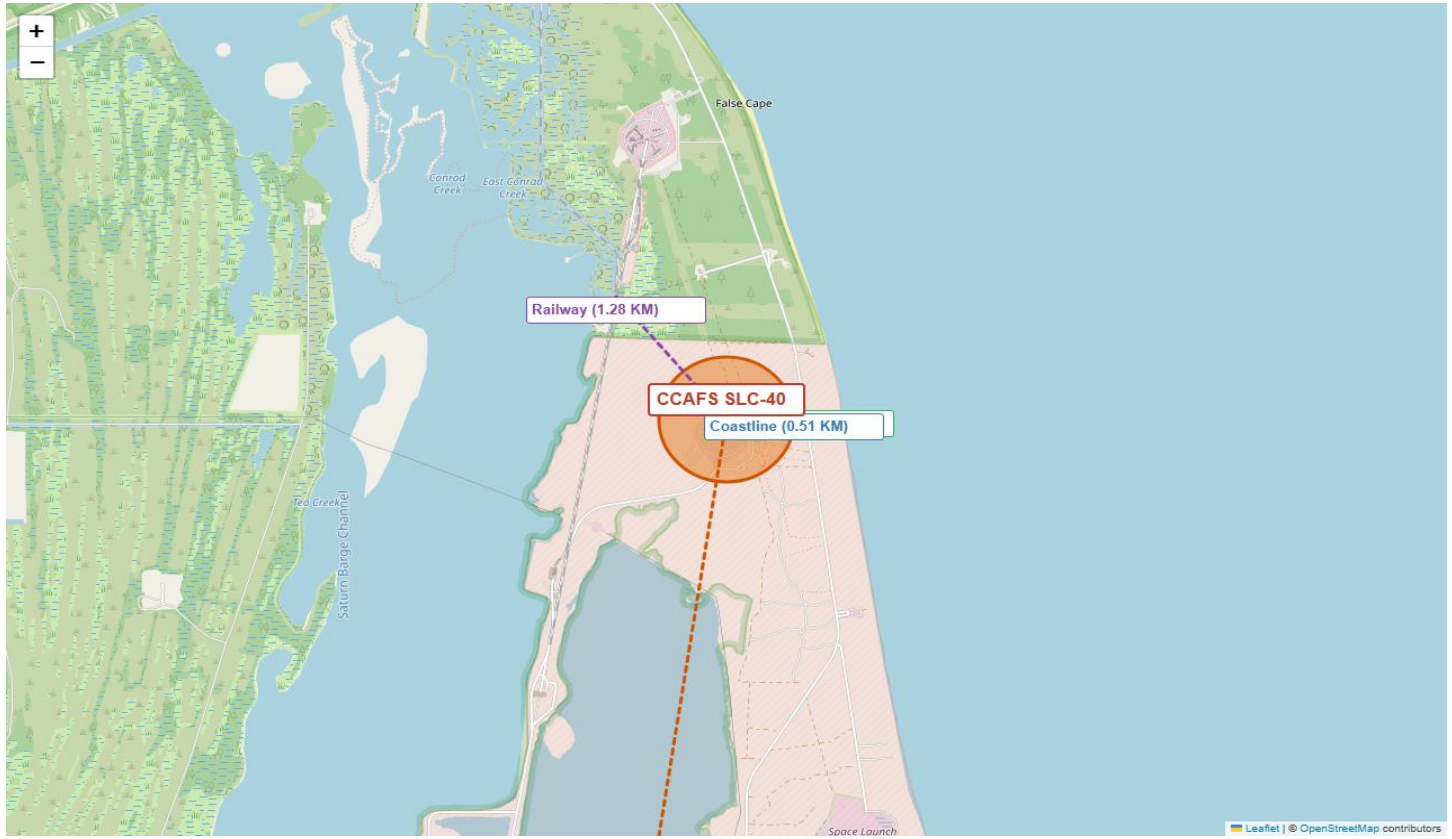
2015-12-22

MIN(Date)

Representative SQL queries

Exact F9 v1.1 (2,534.7 kg with sub-variants)

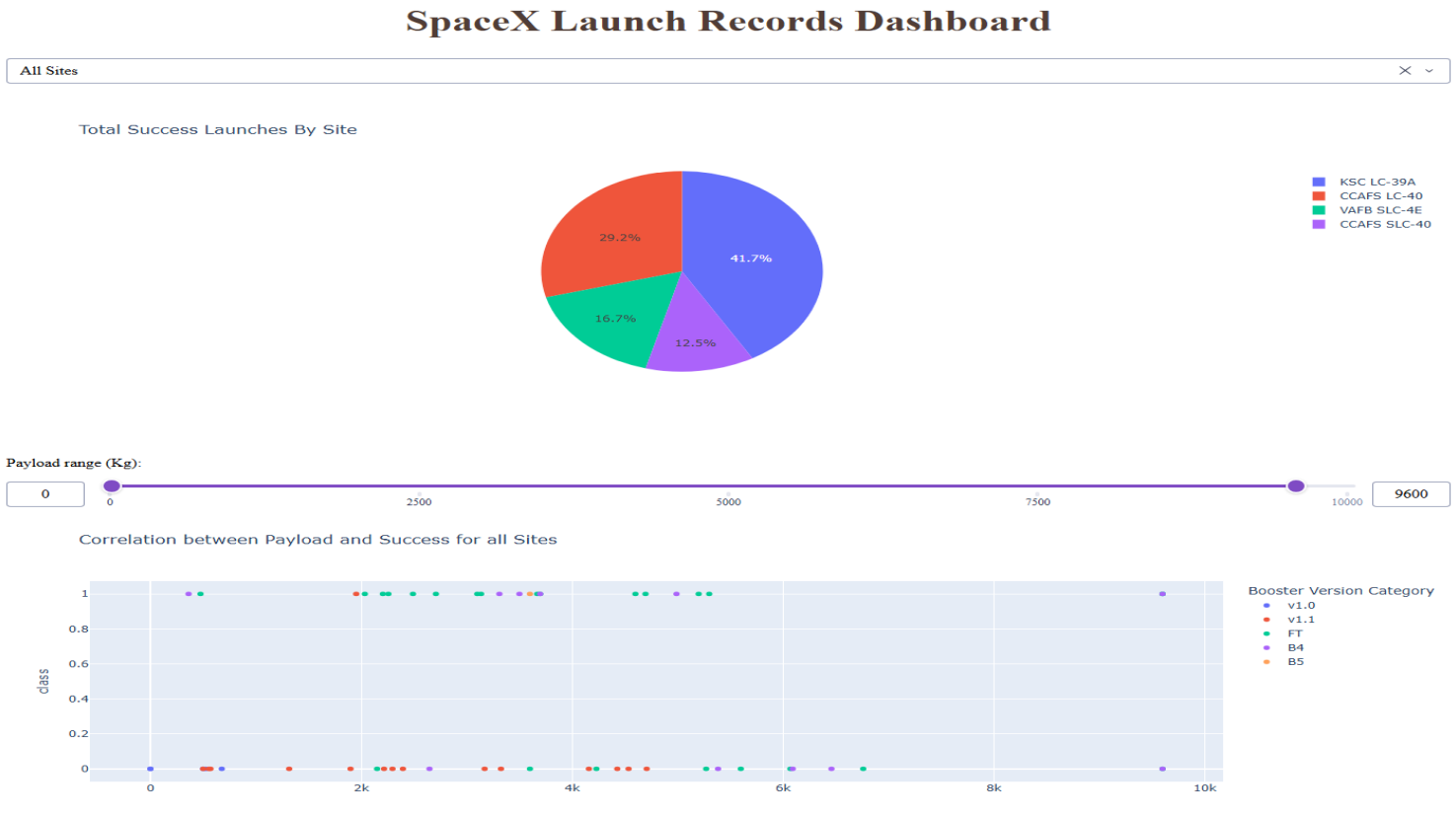
Interactive Map — Folium



- Mapped the three canonical launch locations using latitude and longitude.
- Launch records can be grouped by site and visually inspected for outcome patterns.
- Geospatial analysis highlights how launch infrastructure is concentrated on coastal sites.
- The interactive Folium notebook supports marker popups, clustering and site-level exploration.

Report view shown above; the course activity uses an interactive Folium map.

Interactive Dashboard — Plotly Dash



- Launch-site selector allows site-level comparison.
- Payload range filtering isolates light, medium and heavy missions.
- Dynamic charts connect payload mass, flight number and landing outcome.
- Dashboard converts the notebook analysis into a stakeholder-friendly exploration tool.

Dashboard result view • filters are designed for non-technical stakeholders.

Predictive Analysis — Classification Methodology

Prepare

Encode categorical variables and standardize features

Split

80% training / 20% test hold-out evaluation

Tune

GridSearchCV with 10-fold cross-validation

Compare

Accuracy + confusion matrix / classification metrics

Select

Choose the strongest performing model

Models: Logistic Regression • Support Vector Machine • Decision Tree • K-Nearest Neighbors

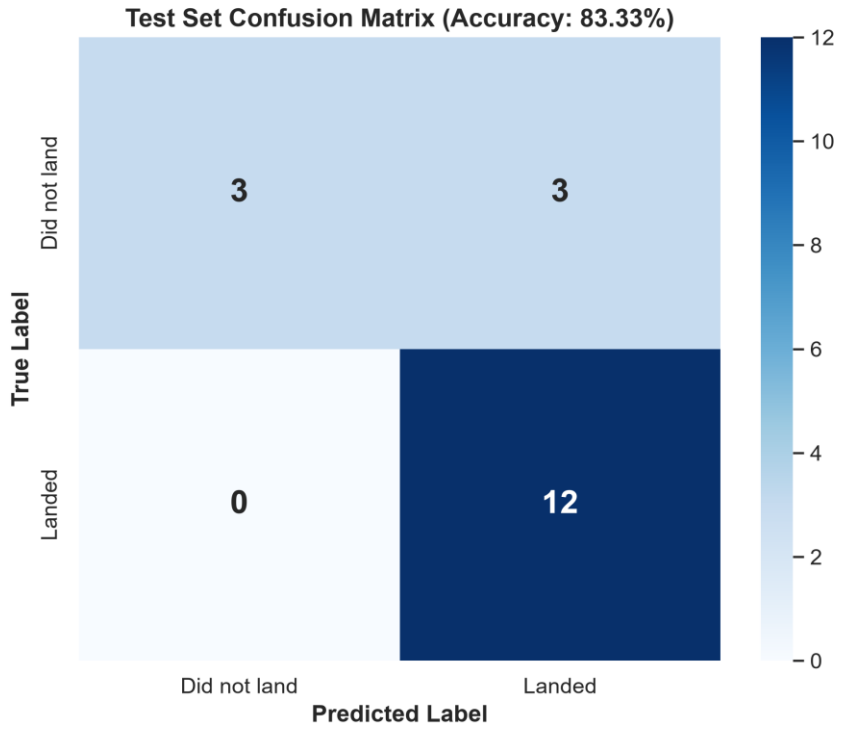
Predictive Analysis — Model Comparison

Logistic Regression	SVM	Decision Tree	KNN
83.3%	**83.3%**	83.3%	**83.3%**
test accuracy	test accuracy	test accuracy	test accuracy

Decision Tree achieved the highest cross-validation score (87.50%).

All four tuned classifiers achieved 83.33% test accuracy (15 of 18 test observations correct).

Best Model — Decision Tree Result



True negatives

5

3

True positives

11

12

False positives

1

3

False negatives

1

0

Interpretation

The model correctly classified 15 of 18 test cases (83.33% accuracy) with zero false negatives (100% recall on successful landings).

Key Insights

1	**Experience matters**	Landing success rises sharply with flight number; later missions show much stronger recovery performance.
2	**Site matters**	KSC LC 39A has the highest success rate among the three canonical sites: 77.3%.
3	**Orbit matters**	VLEO is the most common late-period orbit and shows 85.7% success; GTO is common but lower at 51.9%.
4	**Payload is not a simple penalty**	Heavy payload missions can still land successfully, especially in later reusable-booster operations.
5	**Prediction is feasible**	A tuned Decision Tree reaches 88.9% test accuracy on the course evaluation.

Conclusion & Stakeholder Recommendation

- Historical evidence shows a strong improvement in first-stage recovery success as SpaceX's Falcon 9 program matured.
- Launch site, orbit type, payload profile and flight experience provide useful signals for recovery prediction.
- Interactive mapping and dashboarding make the findings accessible to non-technical stakeholders.
- The Decision Tree was the best-performing classifier in the evaluated model set, leading cross-validation at 87.50% and all four models achieving 83.33% test accuracy.
- For a competing launch provider, a landing-probability estimate can support more informed bid pricing and launch-cost assumptions.

Final takeaway: recovery is increasingly predictable from mission characteristics.

Appendix — Deliverables & Sources

Expected project repository contents

- Data Collection API notebook
- Web Scraping notebook
- Data Wrangling notebook
- EDA with Visualization notebook
- EDA with SQL notebook
- Folium notebook / HTML output
- Plotly Dash Python application
- Predictive Analysis notebook
- Dataset CSV files and README

Repository link

https://github.com/Abhishek5472/IBM-Applied-Data-Science-Capstone

Sources

- IBM / Coursera Applied Data Science Capstone course materials and template.
- SpaceX public REST API launch records.
- Wikipedia Falcon 9 launch-history table used for web-scraping methodology.
- Course-standard 90-record Falcon 9 analysis dataset and associated lab workflow.